

Module 4: Web Application Security



Topic 4: Javascript



What is Javascript?



- **Scripting language** that helps in creating dynamic and interactive websites.
- Used to create **animations**, control **multimedia** and many other tasks
- More than **98%** of modern websites use Javascript within its websites.

What is Javascript?

- In the pentesting field, sometimes you need to have a good knowledge about **javascript** to be able to analyze the web application code and figure out one of the client side vulnerabilities that relies on javascript **(e.g XSS ,CSRF)**.
- You can write the javascript code inside HTML page in <script> tags or you can write it on external js file and import it in the html file using :

```
<script>alert('XSS')</script>
```

```
<script src=youreexternalfile.js></script>
```

Javascript Basics

- Define

```
<script>
    document.write('website is at : ' + window.location.hostname +
</script>
```

- Defining Variable (Global, Local)

```
var _0xae5b = ["\x76\x61\x6c\x75\x65", "\x75\x73\x65\x72", "\x67\x65\x74\x45\x6c\x65\x6d\x65\x6e\x74\x42\x79\x49\x64"
function check() {
    var _0xeb80x2 = document[_0xae5b[2]](_0xae5b[1])(_0xae5b[0]);
    var _0xeb80x3 = document[_0xae5b[2]](_0xae5b[3])(_0xae5b[0]);
    if (_0xeb80x2 == _0xae5b[4] && _0xeb80x3 == _0xae5b[4]) {
```

- Conditions

```
if (hour < 18) {
    greeting = "Good day";
} else {
    greeting = "Good evening";
}
```


Javascript Basics

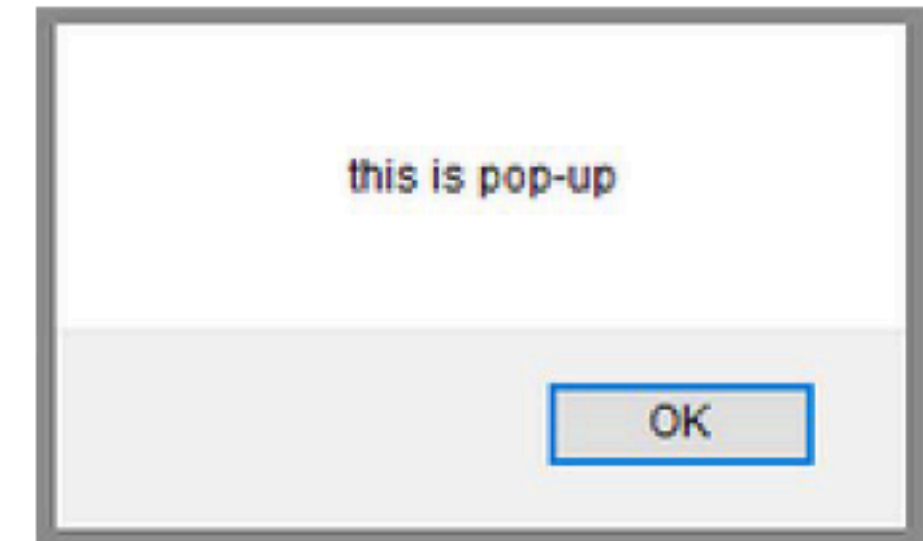
- **Printing**

- Document (DOM)

```
function myFunction() {  
    document.write("Hello World!");  
}
```

- Popup

```
alert('this is pop-up');
```



- Browser Console

```
console.log('write in the console');
```

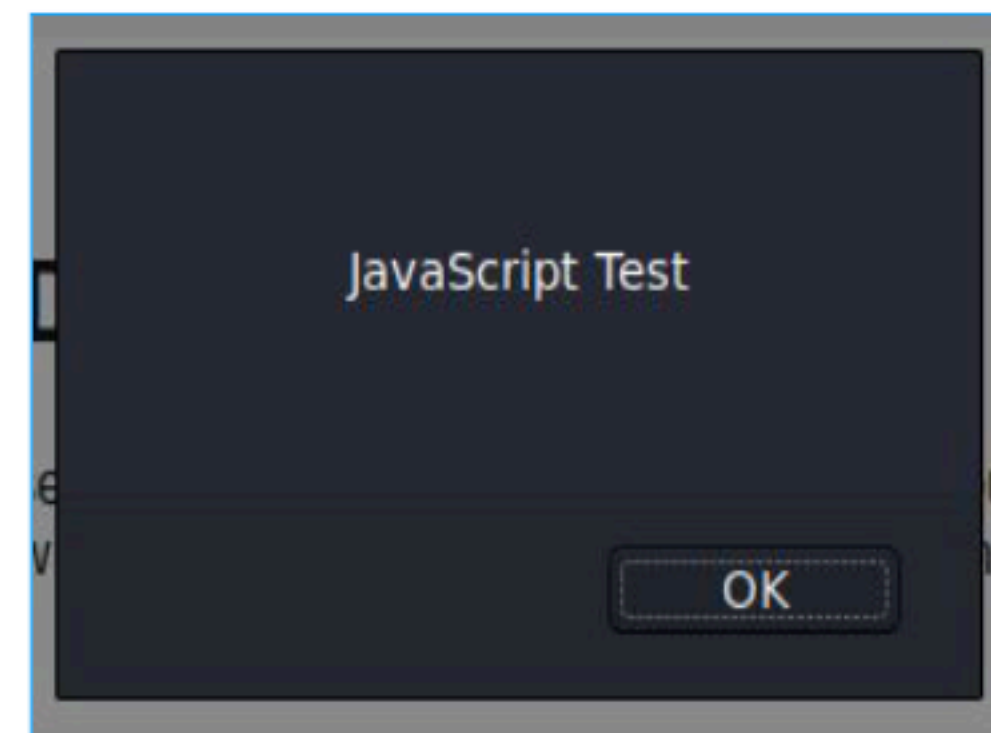
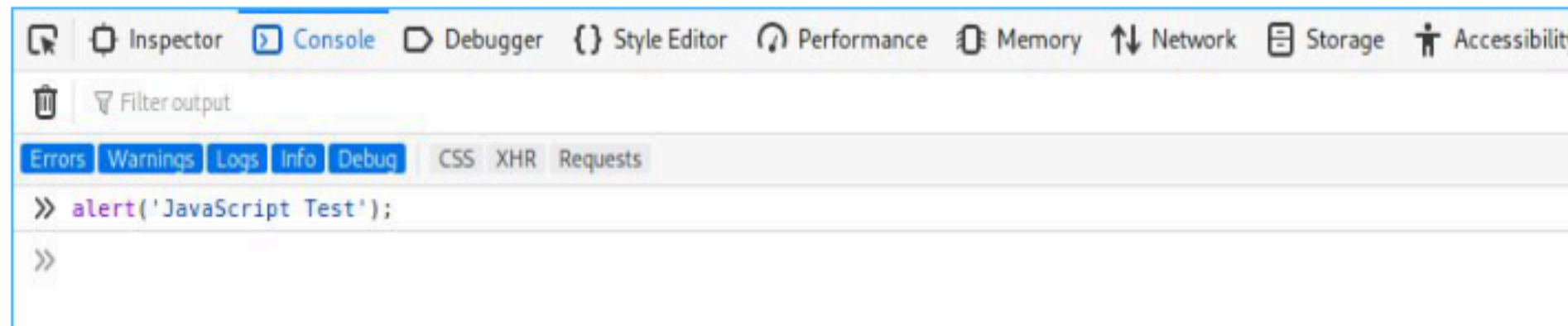

Javascript Basics

- Reading Params

```
url = location.search; // e.g. ?num1=43&num2=23  
var parts = url.substring(1).split('&');  
para_str = parts[0];
```

How to Run Javascript?

- You can execute JS code using webdev tools on your browser (Right click + inspect)



Lab

This is Sparta

